

Unit 1

Solutions - PART 1

MULTIPLE CHOICE QUESTIONS

1. The atmospheric pollution is generally measured in the units of:
(a) mass percentage (b) volume percentage
(c) volume fraction (d) ppm

2. 200 gm of sugar syrup contains 34.2 gm of sugar. What is the molal concentration of the solution?
(a) 0.1 (b) 0.5
(c) 5.5 (d) 55

3. Among the following substance the lowest vapour pressure is exerted by:
(a) water (b) alcohol
(c) ether (d) mercury

4. The law which indicates the relationship between solubility of a gas in liquid and pressure is:
(a) Raoult's law (b) Henry's law
(c) Lowering of vapour pressure (d) Van't Hoff law

5. Partial pressure of a solution component is directly proportional to its mole fraction. This is known as:
(a) Henry's law (b) Raoult's law
(c) Distribution law (d) Ostwald's dilution law

6. Which of the following solutions shows positive deviation from Raoult's law?
(a) Acetone + Aniline (b) Acetone + Ethanol
(c) Water + Nitric acid (d) Chloroform + Benzene

7. The system that forms maximum boiling azeotrope is:
(a) Acetone-chloroform (b) ethanol-acetone
(c) n-hexane-n-heptane (d) carbon disulphide-acetone

8. The relative lowering in vapour pressure is proportional to the ratio of number of:
- (a) solute molecules to solvent molecules
 - (b) solvent molecules to solute molecules
 - (c) solute molecules to the total number of molecules in solution
 - (d) solvent molecules to the total number of molecules in solution
9. Which of the following units is useful in relating concentration of solution with its vapour pressure?
- (a) Mole fraction
 - (b) Parts per million
 - (c) Mass percentage
 - (d) Molality
10. At equilibrium the rate of dissociation of a solid solute in a volatile liquid solvent is
- (a) less than the rate of crystallization
 - (b) greater than the rate of crystallisation
 - (c) equal to the rate of crystallisation
 - (d) zero
11. Temperature dependent term among the following is –
- (a) Molality
 - (b) Molarity
 - (c) Mole fraction
 - (d) Weight percentage
12. Camphor in nitrogen gas, is an example of ____.
- a) liquid Solutions
 - b) solid Solutions
 - c) gaseous Solutions
 - d) aqueous solution
13. Out of following which one is not an example of a solution.
- a) Air
 - b) Brass
 - c) amalgam
 - d) benzene in water

14. Match the following Type solution with their examples:

Type of solution	Example
A. Gaseous solution	p. dissolved oxygen in drinking water
B. Liquid solution	q. amalgam of mercury with sodium
C. Solid solution	r. atmospheric air
	s. Silicon

- a) A – q; B – r; C – p b) A – r; B – p; C – q
 c) A – q; B – s; C – p d) A – s; B – r; C – q

15. Match the items given in Column I with the type of solutions given In Column II.

	Column I		Column II
(i)	Soda water	A.	A solution of liquid in solid
(ii)	Sugar solution	B.	A solution of gas in gas
(iii)	German silver	C.	A solution of solid in liquid
(iv)	Air	D.	A solution of solid in solid
		E.	A solution of gas in liquid

- a. (i) - B; (ii) - C; (iii) – D; (iv) – E; b) (i) - E; (ii) - C; (iii) – D; (iv) – B;
 c) (i) - B; (ii) - E; (iii) – D; (iv) – B; d) (i) - E; (ii) - C; (iii) – B; (iv) – D;

16. The concentration term depends on temperature is

- a. ppm b) mole fraction
 c) molality d) molarity

18. Which of the following is dependent on temperature?

- a. Molarity b) Mole fraction
 c) Weight percentage d) Molality

19. Increasing the temperature of an aqueous solution will cause

- a) decrease in molality b) decrease in molarity
 c) decrease in mole fraction d) decrease in mass percentage (w/w)

20. The mass percentage (w/w) of glucose in water is 10% means
- 10g of glucose dissolved in 100g of water
 - 10g of glucose dissolved in 10g of water
 - 10g of glucose dissolved in 90g of water
 - 10g of glucose dissolved in 90ml of water
21. The mass/volume percentage(w/v) of glucose in water is 10% means.
- 10g of glucose dissolved in 100 g of water
 - 10g of glucose dissolved in 90 mL of water
 - 10g of glucose dissolved in 100 mL of water
 - 10g of glucose dissolved in 90 g of water
22. The mole fraction of the solute in one molal aqueous solution is
- 0.009
 - 0.018
 - 0.027
 - 0.036
23. If one mole of a substance is present in one kg of solvent, then
- It shows molar concentration
 - It shows molal concentration
 - It shows normality
 - It shows strength g/g
24. The concentration of fluoride ions to be present in water to prevent tooth decay is
- 1 ppm
 - 1.5 ppm
 - 2 ppm
 - 2.5 ppm
25. The concentration of pollutants in water or atmosphere is expressed in terms of
- $\mu\text{g L}^{-1}$
 - molarity
 - molality
 - volume percentage (v/v)
26. The concentration of fluoride ions to be present in water to prevent tooth decay is
- 1 ppm
 - 1.5 ppm
 - 2 ppm
 - 2.5 ppm

27. The tanks used by most of scuba divers are filled with air diluted with helium of around

- a. 88.3%
- b) 56.2%
- c) 32.1%
- d) 11.7%

28. A student took two glasses of pure water from a water filter. He cools glass-A in a fridge and warms the other glass – B on a stove. On comparing the solubility of oxygen in H_2O in glass-A and glass-B, he states that glass-‘A’ contains

- a) more oxygen than glass-B.
- b) less oxygen than glass-B.
- c) same amount of oxygen as in glass-B.
- d) zero concentration of oxygen.

29. The dissolution of gas in a liquid is governed

- a) Raoult's law
- b) Henry's law
- c) Dalton's law of pressure
- d) van't Hoff factor

30. P_A and P_B are the vapour pressures of pure liquid components, A and B, respectively of an ideal binary solution. If X_A represents the mole fraction of component A, the total pressure of the solution will be

- a. $P_A + X_A (P_B - P_A)$
- b) $P_A + X_A (P_A - P_B)$
- c) $P_B + X_A (P_B - P_A)$
- d) $P_B + X_A (P_A - P_B)$

31. The vapour pressure of two liquids P and Q are 80 and 60 torr, respectively. The total vapour pressure of solution obtained by mixing 3 mole of P and 2 mole of Q would be

- | | |
|------------|-------------|
| b. 72 torr | b) 140 torr |
| c) 68 torr | d) 20 torr |

32. Which of the following statements about the composition of the vapour over an ideal 1: 1 molar mixture of benzene and toluene is correct? Assume that the temperature is constant at 25°C. (Given, vapour pressure data at 25°C, benzene 12.8kPa, toluene=3.85kPa).

- c. The vapour will contain equal amounts of benzene and toluene.
- d. Not enough information is given to make a prediction.
- e. The vapour will contain a higher percentage of benzene.
- f. The vapour will contain a higher percentage of toluene.

33. The mixture which shows positive deviation from Raoult's Law is

- | | |
|---------------------------|---------------------------------|
| a) n-hexane and n-heptane | b) Bromoethane and chloroethane |
| c) Ethanol and Acetone | d) Chloroform and Acetone |

34. The mixture which shows positive deviation from Raoult's law is

- | | |
|-------------------------|--------------------------------|
| a) ethanol + acetone | b) benzene + toluene |
| c) acetone + chloroform | d) chloroethane + bromoethane. |

35. The mixture that forms maximum boiling azeotrope is

- | | |
|---------------------|---------------------------------|
| a) heptane + octane | b) water + nitric acid |
| c) ethanol + water | d) acetone + carbon disulphide. |

36. Which of the following statements is correct regarding a solution of two components A and B exhibiting positive deviation from ideal behaviour?

- a. Intermolecular attractive forces between A-A and B-B are stronger than those between A-B.
- b. $\Delta V_{\text{mix}}H = 0$ at constant T and P.
- c. $\Delta V_{\text{mix}}V = 0$ at constant T and P.
- d. Intermolecular attractive forces between A-A and B-B are equal to those between A-B.

37. A solution containing components A and B follows Raoult's law.

- a. A-B attraction force is greater than A - A and B-B
- b. A-B attraction force is less than A - A and B - B
- c. A-B attraction force remains same as A - A and B-B
- d. volume of solution is different from sum of volume of solute and solvent.

38. All form ideal solution except

- | | |
|--|---|
| a. C_6H_6 and $\text{C}_6\text{H}_5\text{CH}_3$ | b) C_2H_6 and $\text{C}_2\text{H}_5\text{I}$ |
| c) $\text{C}_6\text{H}_5\text{Cl}$ and $\text{C}_6\text{H}_5\text{Br}$ | d) $\text{C}_2\text{H}_5\text{I}$ and $\text{C}_2\text{H}_5\text{OH}$ |

39. The mixture which shows positive deviation from Raoult's Law is

- | | |
|---------------------------|---------------------------------|
| a) n-hexane and n-heptane | b) Bromoethane and chloroethane |
| c) Ethanol and Acetone | d) Chloroform and Acetone |

40. 10 ml of liquid 'A' and 20 ml of liquid 'B' are mixed at 25°C. The volume of the solution was measured to be 30.1 ml then,

- $\Delta H_{\text{mix}} > 0$, Solution shows negative deviation from Raoult's law.
- $\Delta H_{\text{mix}} < 0$, Solution shows negative deviation from Raoult's law.
- $\Delta H_{\text{mix}} > 0$, Solution shows positive deviation from Raoult's law.
- $\Delta H_{\text{mix}} < 0$, Solution shows positive deviation from Raoult's law.

41. The mixture that forms maximum boiling azeotrope at specific concentration is

- heptane + octane
- water + Nitric acid
- ethanol + water
- acetone + carbon disulphide

42. Match the following Type of liquid - liquid solution with their examples:

	Type of liquid - liquid solution		Example
A.	Ideal solution	p.	ethanol - acetone
B.	Non ideal solution with positive deviation	q.	acetone - chloroform
C.	Non ideal solution with negative deviation	r.	benzene – toluene

- A - p; B - q; C - r
- A - p; B - r; C - q;
- A - q; B - r; C - p;
- A - r; B - p; C - q;

43. In a solution containing non-volatile solute, the mole fraction of solvent is 0.9. The relative lowering of vapour pressure is

- 1
- 0.1
- 0.9
- 1.1

